

MSTCONNECT

FULL STACK WEB DEVELOPMENT INTENSIVE BOOTCAMP

Minimum Computer / Machine Specification

1. Purpose

This document defines the minimum and recommended computer specifications for participants of the MSTCONNECT PH Full-Stack Web Development Bootcamp. The requirements are intended to support HTML, CSS, JavaScript, Git/GitHub, Node.js, Express, MongoDB, Postman, React, and full-stack project development throughout the intensive training.

2. Minimum and Recommended Specifications

Component	Minimum Requirement	Recommended
Processor	Intel Core i3 8th Gen / AMD Ryzen 3 equivalent	Intel Core i5 10th Gen+ / AMD Ryzen 5 4000+
CPU	2 cores / 4 threads	4-6 cores or better
RAM	8 GB	16 GB
Storage	128 GB SSD with 30-40 GB free	256-512 GB SSD
Display	1366 x 768	1920 x 1080 Full HD
Operating System	Windows 10 64-bit, macOS, or modern Linux	Windows 11 64-bit / current supported macOS
Internet	Stable 5 Mbps connection	10-25+ Mbps
Graphics	Integrated graphics	Integrated graphics is sufficient
Keyboard / Mouse	Keyboard and pointing device required	External mouse recommended
Webcam / Microphone	Optional for onsite classes	Recommended for online/hybrid activities

Important: A dedicated graphics card (GPU) is not required. This bootcamp focuses on web application development rather than graphics rendering, video editing, game development, or local AI-model training.

3. RAM and Storage Guidance

The official minimum should be 8 GB RAM with an SSD. A 16 GB RAM configuration is strongly recommended because students may simultaneously run Visual Studio Code, several browser tabs, Node.js/Express, a React development server, Postman, database tools, and the operating system.

Configuration	Suitability
4 GB RAM + HDD	Not recommended
4 GB RAM + SSD	Suitable only for basic early lessons; not recommended for the full bootcamp
8 GB RAM + SSD	Minimum acceptable
16 GB RAM + SSD	Recommended
32 GB RAM	Excellent, but unnecessary for most participants

4. Required Software

- Visual Studio Code - primary code editor
- Git - version control
- GitHub account - repositories and portfolio management
- Node.js and npm - JavaScript runtime and package management
- Google Chrome, Microsoft Edge, or Firefox - browser and developer tools
- Postman - REST API testing
- MongoDB Atlas - recommended cloud database service
- React with Vite - front-end application development
- Terminal / PowerShell
- Bootcamp communication platform, if required (e.g., Google Meet, Zoom, or Discord)

Recommendation: Use MongoDB Atlas. This reduces the resource load on minimum-specification computers.

5. Official Procurement / Participant Specification

Minimum Acceptable Specification

Intel Core i3 8th Generation / AMD Ryzen 3 or better; 8 GB RAM; 256 GB SSD; 64-bit Windows 10/11, current macOS, or a modern Linux distribution; Wi-Fi; and a 1366 x 768 or higher display.

Preferred Specification

Intel Core i5 10th Generation / AMD Ryzen 5 4000 Series or better; 16 GB RAM; 512 GB SSD; Full HD (1920 x 1080) display; and Windows 11 64-bit or current supported macOS.

Refurbished computers: A properly maintained older Core i5 machine with 8-16 GB RAM and an SSD may be more suitable than a newer low-end Celeron/Pentium-class machine. Processor generation should therefore not be used as the only purchasing criterion.

6. Final Recommendation

For this intensive bootcamp format, **8 GB RAM with SSD storage** should be treated as the hard minimum, while **16 GB RAM with SSD storage** should be the recommended configuration. Adequate machine performance helps ensure that class time is spent on coding and project development rather than troubleshooting slow or underpowered computers.

Installation Checklist

- Google Chrome installed
- Visual Studio Code installed
- Live Server extension installed
- Prettier extension installed
- Git installed
- GitHub account created
- Node.js installed
- npm verified
- Postman installed
- MongoDB Atlas account created
- MongoDB database user created
- MongoDB connection string copied
- Bootcamp workspace folder created

Software Installation Guide

Required Applications

Google Chrome

Used for testing websites and opening Developer Tools.

Visual Studio Code

Used as the main code editor.

Recommended extensions:

- Live Server
- Prettier
- E S Lint
- Auto Rename Tag
- Material Icon Theme

Git

Used for version control.

Verify installation:

```
git --version
```

GitHub Account

Used to store and submit project repositories.

Node.js and npm

Used for backend and React development.

Verify installation:

```
node --version  
npm --version
```

Postman

Used to test backend API endpoints.

MongoDB Atlas

Used as the cloud database for backend projects.

Recommended Workspace

```
lastname - mstconnect - fullstack- bootcamp/
```

```
├─ activities /
```

```
├─ projects /
```

```
├─ capstones /
```

```
└─ notes /
```